

# SANGAMESH GIRISH DANDIN

AI Security Engineer | Adversarial ML | LLM Security | Penetration Testing

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## PROFESSIONAL SUMMARY

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AI Security Engineer focused on adversarial machine learning and LLM security. Builds and deploys production-ready systems including prompt injection defense pipelines, AI-driven red team tools, and explainable malware classification models. Published IEEE conference author with strong experience in end-to-end system design, from model development to full-stack deployment. Focus on securing AI systems and applying AI to real-world cybersecurity problems.

## TECHNICAL SKILLS

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**Languages:** Python, JavaScript, Bash, SQL

**AI & ML:** Machine Learning, Deep Learning, NLP, LightGBM, XGBoost, SHAP, LSTM, Random Forest, Gradient Boosting

**Frameworks:** FastAPI, Flask, React, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, ReportLab

**Cybersecurity:** Prompt Injection Defence, Adversarial ML, Penetration Testing, CVE/NVD Analysis, CVSS, Nmap, Metasploit, Burp Suite, Wireshark, Kali Linux, Shodan, OSINT

**Systems & Cloud:** Docker, Docker Compose, AWS (EC2, S3, IAM, VPC), Vercel, Supabase, REST APIs, WebSockets, Git

## PROJECTS

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### [ZeroInject Shield — Prompt Injection Defence System](#)

*Python · FastAPI · Groq LLaMA · React · SQLite · Docker*

- Built a 6-stage zero-trust middleware pipeline - sanitize, verify, consensus, policy-check, enforce, audit blocking prompt injection, jailbreaks, and mixed-intent attacks before they reach the LLM.
- Deployed multi-agent consensus engine across parallel Groq LLM verifiers; benchmarked against JailbreakBench with real-time SAFE / FLAGGED / BLOCKED telemetry dashboard.
- Engineered intent-preservation logic that sanitises ambiguous prompts instead of blocking them reducing false positives while maintaining full security coverage.

### [ByteHunter — AI Malware Classification Engine](#)

*Python · FastAPI · LightGBM · XGBoost · SHAP · LIEF · React · Docker Compose*

- Built a 4-model ensemble (LightGBM + XGBoost + RF + LR) for Windows PE malware classification using a 2,381-dimensional EMBER feature vector extracted via LIEF.
- Integrated SHAP TreeExplainer to surface top prediction drivers section entropy, import table, entry point making every verdict interpretable for analysts.
- Ships with batch scanning (5 files), SHA-256 caching, Win32 API behaviour simulation, PDF forensic report export, and one-command Docker Compose deployment.

### [RedTeam Copilot — AI-Powered Penetration Testing Assistant](#)

*Python · FastAPI · Groq LLM · React · Nmap · Shodan · NVD API · Supabase · Vercel*

- Deployed full-stack AI red team assistant live at redteam-copilot.vercel.app automates the full recon-to-report pipeline: OSINT, Nmap port scanning, NVD CVE matching, AI security analysis.
- Streams live scan progress via SSE; generates structured PDF pentest reports with risk matrix, attack chain, and remediation roadmap in one automated workflow.
- Integrated Shodan passive recon and Supabase cloud storage for persistent pentest report archives across sessions.

### [Predictive Patch Scheduler — Vulnerability Management System](#)

*Python · FastAPI · Random Forest · Gradient Boosting · Plotly · WebSockets · SQLite*

- Processed 500+ CVE records; ML models predict optimal patch windows, improving risk prioritisation accuracy by ~35% over manual methods.
- Built CVSS multi-factor risk scoring with exploit-availability and criticality multipliers, plus WebSocket real-time alerting and Plotly analytics suite.

## VulnXploit — Automated Vulnerability Scanner

*Python · Flask · Nmap · JavaScript · Bootstrap 5*

- Automated Nmap scanning across 65,000+ ports with real-time CVE matching and PoC exploit generation cutting manual assessment time by ~60%.
- Produces CSV and PDF scan reports; syntax-highlighted PoC viewer with live statistics dashboard and one-click copy.

## **PUBLICATION**

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### **AlertSphere: Alerting About Disasters and Ensuring Safety Using AI**

Nayana B. P., K. S. Prakruthi, Ananya Rekha Ashok, Sangamesh Girish Dandin, Tanveer Akhlaque Ahmed.  
IEEE ICAMIDA 2025, Aurangabad, India — DOI: 10.1109/ICAMIDA64673.2025.11209472

## **CERTIFICATIONS**

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- **PNPT (Practical Network Penetration Tester) - In Progress** | TCM Security
- **CompTIA Network+ (N10-008)** - Networking fundamentals, protocols, security, and troubleshooting
- **AWS Academy Graduate** - Cloud Architecting (EC2, S3, VPC, IAM, Load Balancing)
- **Tata Group Cybersecurity Analyst Job Simulation** - Threat analysis and incident response

## **EDUCATION**

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**MSc in Artificial Intelligence** | National College of Ireland | Dublin, Ireland | Jan 2026 – Jan 2027 (Expected)

**B.E. in Artificial Intelligence and Data Science** | Nitte Meenakshi Institute of Technology | Bengaluru, India | Feb 2023 – Oct 2025 | 7.49 CGPA / 10

**Diploma in Computer Science and Engineering** | N.V. Polytechnic College | Kalaburagi, India | Aug 2019 – Aug 2022 | 80%

## **LEADERSHIP**

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### **Vice President | Association for Computing Machinery (ACM) | NMIT | Jan 2024 – Jul 2025**

- Led 10+ technical workshops and cybersecurity events, engaging over 200 student participants across departments.
- Managed cross-functional student teams and coordinated with faculty to deliver semester-long technical programs.

### **Executive Committee Member | IEEE NMIT Student Branch | NMIT | Dec 2023 – Jul 2025**

- Organised 6+ research seminars and technical sessions on emerging technology topics including AI and embedded systems.

### **Vice Secretary | IEEE CIS NMIT | NMIT | Nov 2023 – Jul 2025**

- Coordinated AI-focused workshops and student engagement programs, increasing chapter membership by 30%.